

CURRICULUM VITAE

A. Biodata

1	Name (with title)	Gelar Budiman, ST., MT
2	Sex	L
3	Position	Lecturer
4	Employee ID	08780030-1
5	National Lecturer ID	0429057801
6	Place and Birth Date	Bandung, 29 Mei 1978
7	E-mail	gelarbudiman@telkomuniversity.ac.id/ gelar.budiman@gmail.com
8	Mobile	+6281322045924
9	Office Address	Jl. Telekomunikasi Terusan Buahbatu Bandung Indonesia
10	Nomor Telepon/Faks	+62227564108
11	Courses	1. Linear Signal & System
		2. Digital Signal Processing/Discrete Time Signal Processing
		3. Communication System
		4. Computer Vision
		5. Watermarking & Steganography
		6. Coding & Compression
		7. Microprocessor & Interface
		8. Mobile Application
		9. Speech Processing
		10. AI for Wireless Communications
12	Expertise	Classical and Quantum Signal Processing, Watermarking-Steganography, Compressed Sensing, Multimedia System, Audio-Image Signal Processing, Mobile Apps
13	Current Research	Compressed Sensing in Traffic, Image and Audio Watermarking, Quantum Audio/Image Watermarking

B. Educational Background

	Undergraduate	Master	Doctoral
University	ITTelkom	ITTelkom	ITB

Field	Electro-Telecommunications	Electro-Telecommunications	STEI, Electrical Engineering
Start-Finish	1997-2002	2003-2005	2015-2021
Title of Thesis/Dissertation	Semi-Adaptive TMS320C5x Ekualizer implementation	Configuring MIMO MC-CDMA on Rayleigh Fading Canal	Audio Watermarking
Mentor/Promotor Name	Dr. Jangkung Rahardjo	Suhartono, Dr.	Prof. Andriyan

C. Research / Project

No	Years	Research title	Funding source
1	2010	Internal Research : Real Time steganographic implementations of Audio on TMS320c6455	University
2	2010	Internal Research : Adaptive MIMO Configuration on the user's MC-CDMA Multisystem	University
3	2011	Internal Research : Design Build 3D Effet audio using Head-Relation Transfer Function based TMS320c6455 method	University
4	2011	Internal Research : Automation of documentation and scheduling of final task session with SMS Gateway	University
5	2013	Modeling Design WCDMA <i>multiuser</i> with <i>Distributed Space Time Block Code</i> (D-Stbc) in case of cooperative wireless communication in the Fading Canal environment	High Education Ministry
6	2014	Design of modeling MC-CDMA <i>multiuser</i> with <i>Distributed Space Time Block Code</i> (D-Stbc) in case of cooperative wireless communication in the Fading Canal environment	High Education Ministry
7	2014	Internal Research : White Space steganography text with dual compression using LZW-Huffman On Android	University
8	2015	Steganography of Text implementation with DCT that is compressed and encrypted with LZW-RSA on Android (year 1)	High Education Ministry
9	2015	Audio Watermarking with Spread Spectrum Phase 1 Method (October 2014 – January 2015) (International Research Project)	Markany Inc.
10	2015	Multi-Band Spectrum Sensing-based Multiple Testing Procedure on Cognitive Radio	High Education Ministry
11	2015	Audio Watermarking with Spread Spectrum Phase 2 Method (April-June 2015) (International Research Project)	Markany Inc.

12	2015	Internal Research : Analysis of Wavelet compression performance and Slantlet compression on binary image Data, Grayscale and RGB formats BMP, Png and Jpeg	University
13	2016	Image steganography with DCT compressed and encrypted with LZW-RSA on Android (Year 2)	High Education Ministry
14	2016	Internal Research : Adaptive Audio Watermarking system using DWT based on SNR value	University
15	2017	Internal Research : MDCT based Audio Watermarking with genetic algorithm	University
16	2018	Dissertation Research : Audio Watermarking Year 1	High Education Ministry
17	2019	Dissertation Research : Audio Watermarking Year 2	High Education Ministry
18	2020	Dissertation Research : Audio Watermarking Year 3	High Education Ministry
19	2019	Internal Research : CS pada sistem Audio Watermarking LWT-DST-QR	University
20	2019	Internal Research : Hybrid Audio Watermarking Berbasis DWT-DCT-Cepstrum Menggunakan Algoritma Penyisipan QIM dan SMM	University
21	2020	Internal Research : Hybrid Audio Watermarking Pada Domain DST-SWT Dengan Penyisipan SMM-CPT-QIM	University
22	2020	Internal Research : Audio Watermarking Berbasis DWT-DCT Menggunakan Teknik Multibit Spread Spectrum Pada Raspberry Pi	University
23	2020	Internal Research : Implementasi Audio Watermarking Berbasis DWT Menggunakan SMM Pada Raspberry Pi	University
24	2021	Internal Research : Penyisipan Data Pada Video Menggunakan Teknik Multibit Spread Spectrum Dan Compressive Sampling	University
25	2021	Internal Research : Penyembunyian Data dengan Compressive Sensing dan Differensiator Fraksional	University
26	2021	Internal Research : Perancangan Audio Watermarking berbasis LWT-DCTQR-CPT dengan Metode Hybrid	University
27	2021	Internal Research : Compressive Sensing Pada Watermarking Citra Menggunakan Spread Spectrum	University

28	2021	PK YPT : Penerapan Compressive Sensing Pada Audio Watermarking Untuk Pengamanan Konten Dan Efisiensi Data	University
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D. Publication

No	Title	Journal name	Volume/number/year
1	“MIMO MC-CDMA system configuration on Rayleigh Fading Canal”	Telecommunication Journal of IT Telkom December 2007	Volume-12 Number 2 Page. 82-88 ISSN: No. 1410-7066, 2007.
2	“Adaptive MIMO Design In Multiuser MC-CDMA System”	International Conference On Telecommunication October 2010	Page 119-122 ISSN: 1858-2982/2010
3	“The performance analysis of WCDMA-diversities Relay on Fading Canals”	National Conference of ICT-M Polytechnic of Telkom	2011 ISSN: 2088-8252
4	“ The MC CDMA Performance On Single Relay Cooperative System By Diversity Technique In Rayleigh Fading Channel”	NETCOM 2014	2014 ISSN:2200 – 0011
5	“The Performance Analysis Of Multi-User WCDMA Systems Using D-STBC In Rayleigh Fading Channel”	ICACT 2014	2014 ISSN 1738-9445
6	“Multi User Performance On MC CDMA Single Relay Cooperative System By Distributed STBC In Rayleigh Fading Channel”	IJCNC 2015	Vol. 7 No. 1, January 2015 ISSN: 0974 – 9322
7	“White Space Steganography On Text By Using LZW-Huffman Double Compression”	IJCNC 2015	Vol. 7 No. 2 March 2015 ISSN: 0974 – 9322
8	“Multiple Testing Procedure Based on Energy Detector for Multiband Spectrum Sensing in Cognitive Radio”	APWIMOB 2015	

9	“ The Distributed FFR as the Novelty Solution of the Integration Femtocell and Macrocell in Cellular Network”	APWIMOB 2015	ISBN: 978-1-4799-8290-5
10	“A multicarrier modulation Audio watermarking System”	ICEEI 2015	ISSN 2085 – 6830
11	“Designing Secured Data Using a Combination of LZW Compression, RSA Encryption, and DCT Steganography”	ICWT 2015	ISBN: 978-1-4673-8434-6
12	“Data insertion capacity and complexity with LZW compression and RSA Encryption”	SENASSET 2015	ISBN: 978-602-73672-0-3
13	P-Value Based Cooperative Multiband Spectrum Sensing for Cognitive Radio	ICWT 2015	ISBN: 978-1-4673-8434-6
14	Adaptive Audio Watermarking based on SNR value in Audio File with text insertion information	National Seminar on ITN Seniati Malang (2016)	ISSN: 2085-4218
15	Design Codec-based H. 264 compression algorithm for Video conferencing application	JSM (journal SIFO Mikroskil) 2016	ISSN: 1412-0100
16	Implementation of Watermarking techniques using FFT and Spread Spectrum Watermark on Digital Audio Data	TELKOMIKA 2016	ISSN (P): 2338-8323 ISSN €: 2459-9638
17	Optimization of Discrete Cosine Transform-Based Image Watermarking by Genetics Algorithm	Indonesian Journal of Electrical Engineering and Computer Science 2016	P-ISSN: 2502-4752, E-ISSN: 2502-4760
18	Genetics Algorithm Optimization of DWT-DCT Based Image Watermarking	International Conference on Science and Applied Science (ICSAS) 2016	
19	An Advanced Algorithm To Select Only One Binary Image From RGB	Far East Journal of Electronics and Communications (2016)	ISSN: 0973-7006
20	Audio Watermarking optimization using	National Seminar on applied Technology 2016	19 November 2016

	genetic algorithm with DCT-based Qim method & LWT		
21	Designing and analyzing Audio Watermarking with modified Spectrum Spread and optimized using genetic algorithm	National Seminar on applied Technology 2016	19 November 2016
22	Optimization of Audio Watermarking based on Reduced-ARC BPSK technique using genetic algorithm	National Seminar on applied Technology 2016	19 November 2016
23	A Modified Multicarrier Modulation of Binary Data Embedding in Audio files	International Journal on Electrical Engineering and Informatics (IJEI)	Vol. 4 Dec 2016 P-ISSN 2085-6830/E-ISSN 2087-5886
24	Design of OFDM based Audio Watermarking using method Quantization Index Modulation (QIM) and BCH Code	National Seminar on innovation and technology application in industry 2017	February 4, 2017, ITN Malang, ISSN: 2085-4218
25	Audio Watermarking with Discrete Wavelet Transform and Histogram using genetic algorithm optimization	National Seminar on innovation and technology application in industry 2017	February 4, 2017, ITN Malang, ISSN: 2085-4218
26	Optimization of Audio Watermarking based on Log Coordinate Mapping feature (LCM) with Spread Spectrum method using genetic algorithm	National Seminar on innovation and technology application in industry 2017	February 4, 2017, ITN Malang, ISSN: 2085-4218
27	Audio analysis Watermarking using Lifting Wavelet based on statistical characteristics of Sub-Band coefficient by optimization of genetic algorithm	National Seminar on innovation and technology application in industry 2017	February 4, 2017, ITN Malang, ISSN: 2085-4218
28	Designing and analyzing Audio-based Watermarking Digital modulation techniques with convolution encoding	National Seminar on innovation and technology application in industry 2017	February 4, 2017, ITN Malang, ISSN: 2085-4218
29	Analysis of Digital Audio Watermarking based Lifting Wavelet Transform on frequency Domain with Spread Spectrum method	National Seminar on innovation and technology application in industry 2017	February 4, 2017, ITN Malang, ISSN: 2085-4218

30	Optimized Audio Watermarking based on Discrete Cosine Transform with Singular Value Decomposition technique using genetic algorithm	National Seminar on innovation and technology application in industry 2017	February 4, 2017, ITN Malang, ISSN: 2085-4218
31	Designing discrete-based Audio Watermarking Wavelet Transform and Modified Discrete Cosine Transform with genetic algorithm optimization	National Seminar on innovation and technology application in industry 2017	February 4, 2017, ITN Malang, ISSN: 2085-4218
32	Analysis and implementation of the optimization of genetic algorithms on Audio Watermarking using the Discrete algorithm method of Wavelet Transform and Complex Cepstrum Transform	National Seminar on innovation and technology application in industry 2017	February 4, 2017, ITN Malang, ISSN: 2085-4218
33	Optimization Audio Watermarking-based Discrete Wavelet Transform (DWT) using M-Ary technique using genetic algorithm	National Seminar on innovation and technology application in industry 2017	February 4, 2017, ITN Malang, ISSN: 2085-4218
34	Removable Watermarking for Cyber Crime control over Digital Audio	Electronics and Telecommunications journals	P-ISSN: 1411-8289, E-ISSN: 2527-9955
35	Feasibility study of strengthening the public – private partnership for tuberculosis case Detection in Bandung City, Indonesia	BMC Research Notes	Vol 10:404, ISSN, 1756-0500, 2017
36	FFT-Based Audio Watermarking in Adaptive Subband with Spread Spectrum Framework	Proceedings of the International, 2 nd Advanced Research in Electrical and Electronic Engineering Technology (ARIEET), 12-14 December 2017	Converted to journal, ISSN 2180-1843, E-ISSN 2289-8131, 12-14 December 2017
37	Analytical and designing Compressive Sampling by synchronizing on Audio Watermarking Stereo-based Lifting Wavelet Transform with Cepstrum method	National proceedings, national Seminar on Energy Telecommunication and automation 2017, 16 December 2017	Page. 28-1, ISBN: 978 - 602 - 74127 - 4 - 3, 16 Dec 2017

38	Audio Watermarking Stereo with Hybrid-based sync of SWT-Fft-SVD and Qim	National proceeding, National Seminar on Electrical Engineering , 16 December 2017	Page. 15,, 16 December 2017
39	Steganography of SS-based synchronized Audio with combined method LWT-SVD	National proceeding, National Seminar on Electrical Engineering , 16 December 2017	Page. 14,, 16 December 2017
40	Audio Watermarking Stereo synchronized based Stationary Wavelet Transform with Centroid calculations	National proceeding, National Seminar on Electrical Engineering , 16 December 2017	Page. 17,, 16 December 2017
41	Data insertion with combined LWT-DST techniques and Compressive Sampling in steganography of Stereo Audio	National proceeding, National Seminar on Electrical Engineering , 16 December 2017	Hal 7,, 16 December 2017
42	Audio design Watermarking with DWT-Histogram technique applied to WEB application	National proceedings, National Seminar ReTII (engineering and Information Technology) 12 th , 9 December 2017	Page. 61, ISSN: 1907-5995, 9 December 2017
43	Engineering design Cs and synchronization of LWT-based Stereo Watermarking with Hybrid Cepstrum and Histogram methods	National proceedings, National Seminar ReTII (engineering and Information Technology) 12 th , 9 December 2017	Page. 238, ISSN: 1907-5995, 9 December 2017
44	Performance Technique LWT-Dct-Svd on Stereo Watermarking Audio with synchronization and Compressive Sampling	National proceedings, National Seminar ReTII (engineering and Information Technology) 12 th , 9 December 2017	Page. 54, ISSN: 1907-5995, 9 December 2017
45	Design Audio Watermarking Stereo with synchronization using Hybrid DWT and Smm based Fft method	National proceedings, National Seminar ReTII (engineering and Information Technology) 12 th , 9 December 2017	Page. 105, ISSN: 1907-5995, 9 December 2017
46	Combined technical performance Analysis DWT-DST-QR Decomposition on Stereo Watermarking with Compressive Sampling	National proceedings, National Seminar ReTII (engineering and Information Technology) 12 th , 9 December 2017	Page. 69, ISSN: 1907-5995, 9 December 2017
47	Fibonacci Sequence based FFT and DCT performance comparison in Audio Watermarking	International Journal of Engineering and Technology	Vol. 8 No. 19, 2019
48	FFT-Based Data Hiding on Audio	Elektronika Ir Elektrotechnika, 2020	Vol. 26 No. 3, 2020, ISSN : 2029-5731

	in LWT-domain using Spread Spectrum Technique		
49	Compressive Sampling with Multiple Bits Spread Spectrum Based Data Hiding	Applied Science, 2020	Vol. 10 No. 12, 2020, ISSN : 2076-3417
50	Robust audio watermarking based on transform domain and SVD with compressive sampling framework	Telkomnika, 2020	Vol. 18, No. 2, April 2020, ISSN : 1693-6930
51	DWT-SMM-based Audio Steganography with RSA Encryption and Compressive Sampling	Telkomnika, 2020	Vol. 18, No. 2, April 2020, ISSN : 1693-6930
52	Wavelet-Based Hybrid Audio Watermarking Using Statistical Mean Manipulation and Spread Spectrum	27 th International Conference on Telecommunications, 2020	10.1109/ICT49546.2020.9239581
53	A Novel Approach: Effective Compressive Sensing in Power Network Problem	Journal of Hunan University Natural Sciences, 2021	Vol. 47 No. 12

E. Seminar (Oral Presentation)

No	Name of meeting/scientific/Seminar	Scientific article title	Time and place
1	International Conference On Telecommunication October 2010	"Adaptive MIMO Design In Multiuser MC-CDMA System"	October 2010/Savoy Homann Bandung
2	International Conference on Electrical Engineering and Informatics 2015	"A Multicarrier Modulation Audio Watermarking System"	August 2015/Jayakarta Hotel Bali
3	International Conference on Wireless and Telematics 2015	"Designing Secured Data Using a Combination of LZW Compression, RSA Encryption, and DCT Steganography"	November 2015/Quality Hotel Manado
4	International Conference on Science and Applied Science (ICSAS) 2016	"Genetics Algorithm Optimization of DWT-DCT Based Image Watermarking"	19 November 2016/Solo Hotel Syariah
5	The 3rd International Conference on Science, Engineering, Built Environment, and Social Science (ICSEBS 2016)	"Fibonacci Sequence – based FFT and DCT performance comparison in Audio Watermarking"	28 November-1 December 2016/Faculty of Applied Sciences Bandung

6	The 2nd Advanced Research in Electrical and Electronic Engineering Technology (ARIEET 2017)	"FFT-Based Audio Watermarking in Adaptive Subband with Spread Spectrum Framework "	12-14 Dec 2017, Jakarta
7	International Conference on Information and Communication Technology (ICOICT) 2018	"QIM-based Audio Watermarking with Combined Techniques of SWT-DST-QR-CPT Using SSbased Synchronization"	3-4 May 2018, Bandung
8	International Conference on Control, Electronics, Renewable Energy, and Communications (ICCEREC 2018)	"CPT-Based Data Hiding in Selected Subband Using Combined Transform and Decomposition Method"	5 Desember 2018, Bandung.
9	International Conference on Control, Electronics, Renewable Energy, and Communications (ICCEREC 2018)	"QR-based Watermarking in Audio Subband Using DCT"	5 December 2018, Bandung.
10	International Conference on Control, Electronics, Renewable Energy, and Communications (ICCEREC 2018)	"Histogram-Based Audio Watermarking with Synchronization in Stationary Audio Subband"	5 December 2018, Bandung.
11	27 th International Conference on Telecommunications	Wavelet-Based Hybrid Audio Watermarking Using Statistical Mean Manipulation and Spread Spectrum	5-7 October 2020, Virtual Conference.

H. Patent/ HKI Publications

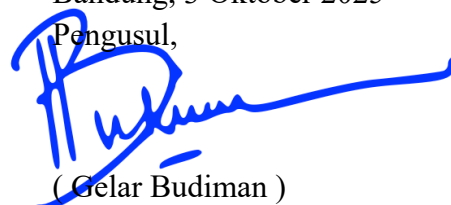
No.	Judul/Tema HKI	Tahun	Jenis	Nomor P/ID
1	Aplikasi Stegacitra	2016	Program Komputer	No. 086109, Kemenkumham, Dirjen HaKi
2	Stegakompresi : 091082	2017	Program Komputer	Dirjen KI
3	CSTV : 000175914	2019	Program Komputer	Dirjen KI
4	Autermarqss : 000202420	2020	Program Komputer	Dirjen KI
5	APK-CM : 000193936	2020	Program Komputer	Dirjen KI
6	LiW-ACS : 000269358	2021	Program Komputer	Dirjen KI

Semua data yang saya isikan dan tercantum dalam biodata ini adalah benar dan dapat dipertanggungjawabkan secara hukum. Apabila di kemudian hari ternyata dijumpai ketidaksesuaian dengan kenyataan, saya sanggup menerima sanksi.

Demikian biodata ini saya buat dengan sebenarnya untuk memenuhi salah satu persyaratan dalam kegiatan penelitian dan pengabdian masyarakat.

Bandung, 3 Oktober 2025

Pengusul,



(Gelar Budiman)

